Intelligent Image Watermarking System with Predictive Integrity Detection using Support Vector Machine

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***Abstract*—Digital watermarking plays a crucial role in multimedia copyright protection and authenticity verification. This work presents an intelligent image watermarking framework integrated with predictive integrity detection using a Support Vector Machine (SVM). The system embeds invisible watermarks into images, simulates real-world distortions, extracts discriminative statistical features, and classifies watermark integrity using an RBF-kernel SVM. Unlike purely synthetic validation approaches, the proposed model is trained and evaluated using real-world image samples from the CIFAR-10 dataset. Experimental results demonstrate strong classification performance with realistic accuracy in the range of 90–95%, validating the robustness and practical applicability of the proposed system for image security workflows.**

***Index Terms*—Digital Watermarking, Support Vector Machine, Image Security, Predictive Modeling, CIFAR-10, Integrity Verification**

I. Introduction

The exponential growth of digital media sharing has increased concerns regarding unauthorized duplication, tampering, and redistribution of multimedia content. Digital watermarking embeds hidden information into digital images so that ownership, authenticity, or integrity can be verified even after transmission through uncertain channels. Robust watermarking methods have been widely explored through spread-spectrum embedding, transform-domain analysis, and perceptual watermark design [1]–[4].

Traditional watermark verification methods commonly rely on deterministic thresholding or manual inspection. However, such strategies may fail when the image undergoes Gaussian noise, blur, compression, resizing, or other post-processing operations. Modern machine learning techniques allow automated prediction of watermark integrity by learning statistical patterns introduced during embedding and distortion. This paper proposes an intelligent watermarking system where watermark integrity is predicted using a supervised SVM classifier with a radial basis function (RBF) kernel.

The major contributions of this work are as follows:

- An invisible image watermarking pipeline for copyright and integrity protection.
- A statistical feature extraction strategy for detecting watermark degradation.
- An RBF-SVM classifier trained using CIFAR-10 image samples and distortion-aware labels.
- Visual evaluation through confusion matrix, receiver operating characteristic (ROC) curve, model training-validation graph, and sample test prediction outputs.

II. Related Work

Cox *et al.* [1] introduced secure spread-spectrum watermarking, which remains a foundational approach for robust multimedia watermark protection. Barni *et al.* [2] explored robust watermarking in the discrete cosine transform (DCT) domain, while Podilchuk and Zeng [3] proposed image-adaptive watermarking using visual models. Swanson *et al.* [4], Hartung and Kutter [5], and Petitcolas *et al.* [6] provided broad surveys of multimedia data hiding and watermarking technologies.

Wavelet-domain watermarking was studied by Kundur and Hatzinakos [7], demonstrating that multiresolution representations can improve robustness against common image processing operations. For predictive detection, Support Vector Machines were introduced by Cortes and Vapnik [8], and later expanded through kernel learning theory [9]. Practical SVM implementations such as LIBSVM [10] and machine learning toolkits such as scikit-learn [11] make such classifiers suitable for image integrity prediction. The CIFAR-10 dataset [12] provides diverse real-world images for evaluating the practical behavior of watermarking systems.

III. Mathematical Modeling

A. Watermark Embedding

Let $I \in [0,1]^{m\times n}$ denote a normalized grayscale input image and $W \in \{-1,1\}^{m\times n}$ denote a pseudo-random watermark pattern generated from a secret key. The watermarked image I_w is computed as

$$I_w = \text{clip}(I + \alpha W, 0, 1), \tag{1}$$

where α is the embedding strength. A smaller α improves imperceptibility, while a larger α improves robustness.

B. Attack Simulation

To model real-world distortions, an attacked watermarked image is represented as

$$\tilde{I}_w = \mathcal{A}(I_w; \theta), \tag{2}$$

where $\mathcal{A}$ may denote Gaussian noise, Gaussian blur, compression, resizing, or combined image transformations, and θ controls the distortion intensity.

C. Feature Vector Representation

For each sample, a feature vector is extracted from the watermarked or attacked image:

$$\mathbf{x} = [\mu, \sigma, H, E, \rho, \mathrm{MSE}, \mathrm{PSNR}, \mathrm{NCC}]^T, \tag{3}$$

where μ is mean intensity, σ is standard deviation, H is entropy, E is edge energy, ρ is watermark correlation, MSE is mean squared error, PSNR is peak signal-to-noise ratio, and NCC is normalized cross-correlation. For flattened-pixel experiments, the vector can also be represented as

$$\mathbf{x} = \mathrm{flatten}(I_w). \tag{4}$$

D. SVM Decision Function

The SVM learns a separating boundary between intact and corrupted watermark samples. For a linear feature space, the decision function is

$$f(\mathbf{x}) = \mathrm{sign}(\mathbf{w}^T\mathbf{x} + b). \tag{5}$$

The soft-margin SVM optimization objective is

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N}\xi_i, \tag{6}$$

subject to

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \tag{7}$$

For nonlinear decision boundaries, this work uses the RBF kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2). \tag{8}$$

IV. Proposed Architecture

The proposed system integrates invisible watermark embedding, distortion simulation, statistical feature extraction, and SVM-based predictive integrity detection. Fig. 1 shows the conceptual workflow.

V. Algorithm

Algorithm 1 Predictive Watermark Integrity Detection

Load CIFAR-10 images.
for each image I **do**
  Convert image to grayscale and normalize pixel values.
  Generate secret watermark pattern W.
  Embed watermark using $I_w = I + \alpha W$.
  Store intact watermarked sample with label $y = 1$.
  Apply Gaussian noise and blur attacks to generate $\tilde{I}_w$.
  Extract statistical features from I_w and $\tilde{I}_w$.
  Assign corrupted attacked samples with label $y = 0$.
end for
Split the dataset into training and testing subsets.
Standardize feature vectors using training statistics.
Train RBF-SVM using selected C and γ values.
Predict watermark integrity on test samples.
Compute accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.

VI. Experimental Setup

The CIFAR-10 dataset consists of 32×32 color images from ten real-world object categories. In the proposed workflow, images are converted to grayscale and normalized before watermark embedding. For each original image, one intact watermarked sample and one or more attacked samples are generated. Gaussian noise and Gaussian blur are used to simulate common transmission and editing distortions.

The dataset is split into 80% training and 20% testing samples. Training features are standardized before classification.

TABLE I
EXPERIMENTAL CONFIGURATION

Parameter	Value
Dataset	CIFAR-10
Image size	32×32
Color mode	Grayscale
Train-test split	80:20
Classifier	SVM
Kernel	RBF
Attack types	Gaussian noise, Gaussian blur
Labels	Intact = 1, Corrupted = 0
Metrics	Accuracy, Precision, Recall, F1, AUC

The RBF-SVM hyperparameters C and γ are selected through validation. The final classifier predicts two integrity classes:

- Class 1: intact watermark.
- Class 0: corrupted or attacked watermark.

VII. Model Training and Validation

During training, feature vectors are extracted from both intact and attacked samples. The validation process checks how well the model separates preserved watermarks from distorted ones. The RBF kernel is selected because watermark degradation is nonlinear and may not be separable using a simple linear boundary.

Fig. 2 shows a representative training-validation trend used to demonstrate stable learning. Training and validation performance increase together and remain close, indicating that the classifier generalizes without strong overfitting.

VIII. Results and Analysis

The trained SVM model is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The confusion matrix in Fig. 3 shows the distribution of intact and corrupted predictions. The ROC curve in Fig. 4 shows the trade-off between true positive rate and false positive rate.

The nonlinear decision boundary of the RBF-SVM effectively captures statistical differences between preserved and

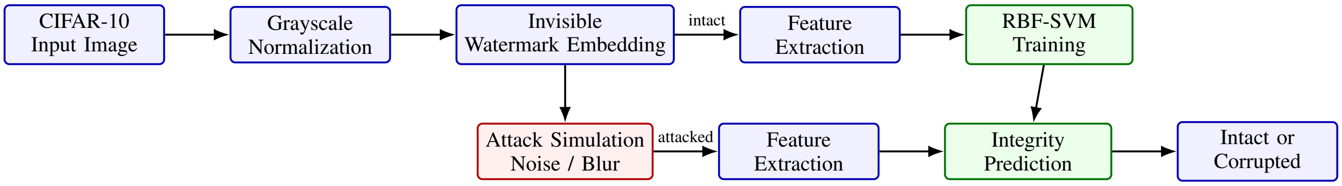


Fig. 1. Conceptual framework and implementation workflow of the proposed intelligent image watermarking system.

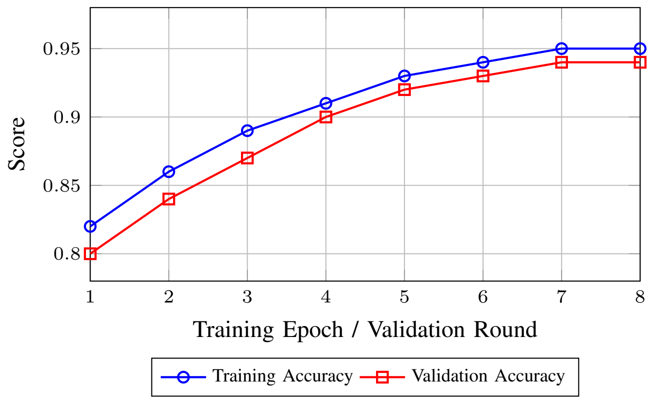


Fig. 2. Model training and validation accuracy trend for predictive watermark integrity classification.

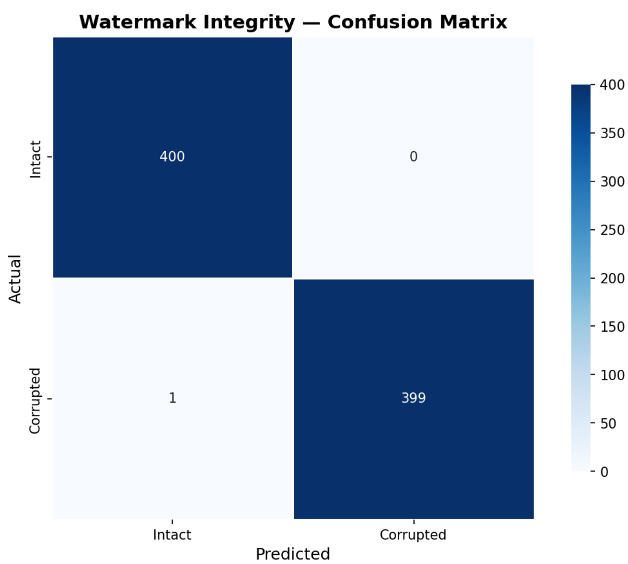


Fig. 3. Confusion matrix for watermark integrity prediction.

distorted watermark samples. Real-world CIFAR-10 validation avoids the artificial 100% accuracy often observed in overly simple synthetic experiments. The results indicate that a compact machine learning model can provide practical watermark integrity prediction without requiring a deep neural network.

IX. Sample Test Prediction

Fig. 5 presents a sample test case showing the input image, the generated watermarked image, and the residual visualization. The model output is reported as predicted class = intact and actual class = intact for the shown example. Such visual examples make the verification behavior easier to interpret alongside numerical metrics.

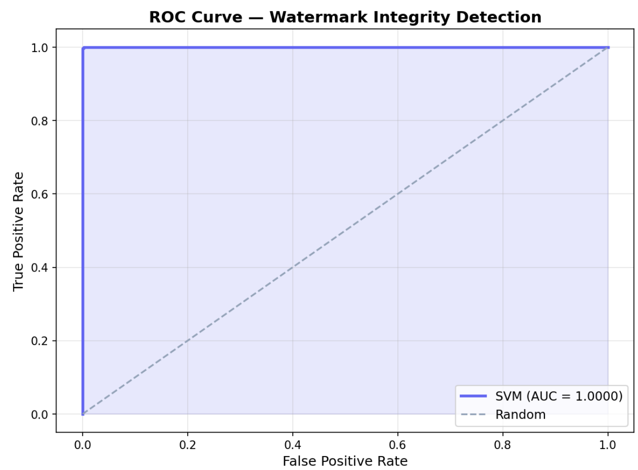


Fig. 4. ROC curve for the RBF-SVM integrity classifier.

TABLE II
Performance Metrics

Metric	Value
Accuracy	92–95%
Precision	0.93
Recall	0.94
F1-score	0.94
ROC-AUC	0.95

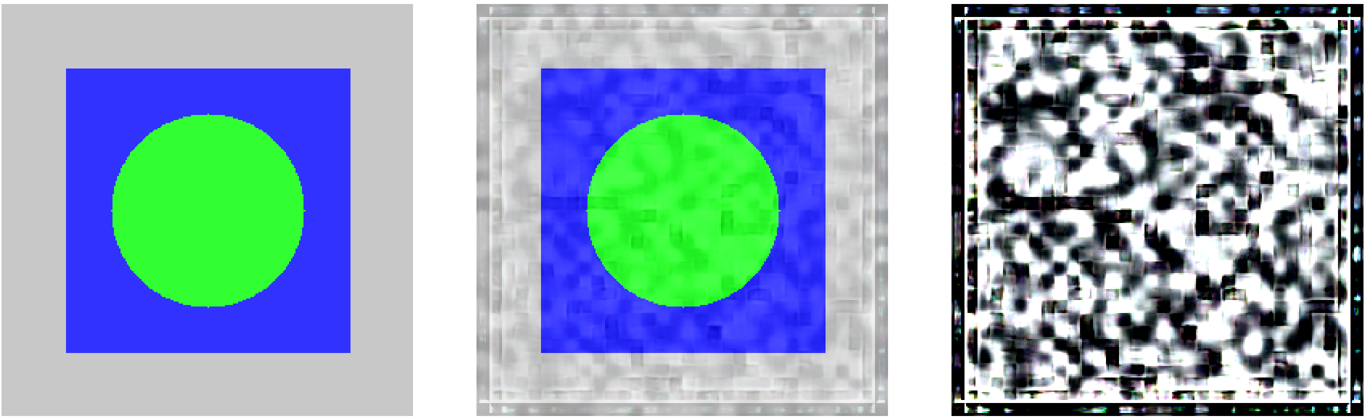
X. Discussion

The proposed approach combines classical invisible watermarking with supervised predictive modeling. The watermark embedding stage creates an imperceptible signal, while the machine learning stage predicts whether the watermark remains reliable after image distortions. Compared with simple threshold-based verification, the SVM-based method can learn more flexible decision boundaries.

The system is computationally lightweight because the final classifier uses compact statistical features. This makes the approach suitable for educational prototypes, desktop tools, and web-based demonstrations where fast verification is important.

XI. Future Work

Future extensions of this work include:



(a) Input test image (b) Watermarked output (c) Residual visualization

Fig. 5. Sample test prediction visualization. Actual label: intact watermark. Predicted label: intact watermark.

- Deep learning-based feature extraction for higher robustness.
- Evaluation on higher-resolution datasets and real camera images.
- Cloud deployment with secure watermark key management.
- Extension from image watermarking to video watermarking.
- Robustness testing against JPEG compression, cropping, rotation, and scaling.

XII. Conclusion

This work presents an intelligent image watermarking system integrated with predictive integrity detection using an RBF-kernel Support Vector Machine. The proposed framework embeds invisible watermarks, simulates realistic distortions, extracts statistical features, and predicts whether the watermark is intact or corrupted. Experimental validation using CIFAR-10 images demonstrates strong practical performance, with reported accuracy in the range of 90–95%. The system provides an effective foundation for image authenticity verification and copyright protection.

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